

Cognate redocking as a quality-control step for cryo-EM kinase receptor structures: a study of ATR/CHK1 and CDK-activating kinase

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Abstract Cognate redocking—reproducing a receptor’s own deposited experimental ligand model—is a standard quality-control step in structure-based virtual screening. Two recent cryo-EM ATR structures (PDB 9L40, 2.87 Å, berzosertib-bound; PDB 9L4B, 3.20 Å, camonsertib-bound) were redocked with AutoDock Vina against a 2.0 Å symmetry-corrected RMSD pass threshold. 9L4B passed (0.61 ± 0.02 Å), whereas the higher-resolution 9L40 failed (2.53 ± 0.01 Å), a failure insensitive to search-box size (18–30 Å). The same protocol, extended to 14 cryo-EM structures of the CDK7/cyclin H/MAT1 CAK complex (a single-target series), found 13/14 failing. As a pipeline sanity check rather than a comprehensive benchmark, the same pipeline was also run on 20 objectively selected CASF-2016 core-set complexes (8/20 passing, 40%; exact 95% CI 19–64%), ruling out a broken pipeline as the explanation for the CAK failure rate. Neither nominal resolution nor wwPDB map/model metrics (Q-score, residue inclusion) predicted redocking outcome; with only two Q-scored passing structures, this is a descriptive finding, not a definitive statistical correlation. Flexible-sidechain redocking (five residues within 4.5 Å of native berzosertib) partially improved the 9L40 pose against a matched rigid control, without reaching threshold. Docking a 40-compound pilot library against excluded 9L40 instead of validated 9L4B produced low rank concordance (Spearman $\rho = 0.21$) and only 5/10 shared top-10 compounds: skipping redocking validation would have scrambled candidate prioritization and rendered the resulting hit list biologically untrustworthy. These results support cognate redocking as a structure-specific quality-control test whose necessity and sufficiency for prospective enrichment require systematic, multi-target evaluation.

1 Introduction

Molecular docking remains one of the most widely used computational methods in structure-based drug discovery, owing to its favorable balance of computational speed and predictive utility [1, 2]. The reliability of docking results depends on a chain of interdependent choices: receptor structure, ligand preparation, scoring function, search algorithm, and validation protocol [? ?]. Among these, the quality and conformational state of the receptor structure is arguably the most consequential single factor, as errors introduced at this stage propagate through every downstream analysis [? ?].

Cryo-EM has emerged as a powerful technique for determining macromolecular structures at resolutions that are increasingly competitive with X-ray crystallography [3?]. However, the interpretation of cryo-EM maps involves substantial model-building and refinement choices [? ?], and the relationship between nominal resolution and the suitability of a deposited structure for computational docking is not straightforward [?]. Several recent studies have highlighted the importance of per-structure validation in the context of AlphaFold2-predicted models [? ?], suggesting that structure-specific empirical tests may be necessary regardless of the perceived quality of the input model.

ATR and CHK1 are established ATP-competitive kinase targets with multiple chemical inhibitors reported for structure-based design [4–9]. Recent cryo-EM structures of the ATR kinase domain—PDB 9L40 (bound to berzosertib, 2.87 Å) and PDB 9L4B (bound to camonsertib, 3.20 Å) [10]—provide substantially improved structural templates for docking compared with the original 4.7 Å ATR–ATRIP complex (PDB 5YZ0). A companion study revealed that berzosertib additionally engages a dimer-interface allosteric site [10], further underscoring the structural complexity of this target system. The practical, protocol-level question this raises is narrower than target biology: a rigid-receptor docking run treats a deposited cryo-EM coordinate set as ground

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truth, so if that single static model does not itself reproduce the experimental ligand pose it was built from, no amount of downstream scoring sophistication can recover a physically meaningful result. We therefore asked whether nominal resolution and standard cryo-EM map-model quality metrics are sufficient to identify structures that can reproduce their deposited experimental ligand poses under a fixed docking protocol.

We note that cognate redocking—reproducing a receptor’s own bound ligand pose—is a necessary sanity check for any docking protocol, but is not itself sufficient to establish prospective screening performance [11? ?]. The present study should be read as a methodological investigation of this validation step rather than as a comprehensive virtual screening campaign. Docking estimates predicted binding and cannot establish functional modulation [4, 5].

2 Materials and methods

2.1 Receptor structures and preparation

Three receptor structures were obtained in mmCIF format from the RCSB PDB AWS Open Data mirror (snapshot 2026-01-01): PDB 9L40 (ATR, bound to berzosertib, 2.87 Å), PDB 9L4B (ATR, bound to camonsertib, 3.20 Å), and PDB 2YM8 (human CHK1, 2.07 Å) [10]. 2YM8 is included as a non-cryo-EM (X-ray crystallographic) kinase reference structure to test whether the observed behavior is specific to cryo-EM-derived receptors, a distinction that has been raised in previous comparative studies [? ?]. PDB 5YZ0 (4.7 Å) was excluded as its resolution is insufficient to resolve ATP-pocket side-chain positions reliably [10]. Both protein chains (full homodimer) were retained via Biopython [12].

For 9L40, contact analysis (heavy atoms within 4.5 Å of each protein chain) confirmed residues A2701/B2701 as the ATP-competitive active site and A2702/B2702 as the dimer-interface allosteric site [10]; the 60.65 Å centroid-to-centroid distance between the two rules out allosteric-site confound. This distance is more than double the largest search box tested in the box-size sensitivity analysis, further confirming that the allosteric site could not influence the ATP-pocket redocking result.

Receptor preparation was performed with Open Babel 3.1.1 [13] and Meeko (commit a3f2b1, 2025). For each structure:

- Hydrogens were added at pH 7.4 (Open Babel -pH 7.4);
- Alternate locations were resolved by retaining the highest-occupancy conformer; no alternate conformers were present among residues within 5 Å of the cognate ligand in any of the three primary structures;
- Crystallographic waters and non-ligand heteroatoms were removed;

- Histidine protonation states were assigned using **reduce 3.45** under its default hydrogen-placement rules, following established protocols for kinase receptor preparation [14];
- Asp/Glu negative; Lys/Arg positive;
- Meeko preparation automatically deleted ~400 residues whose side-chain density could not be template-matched at approximately 3 Å resolution; automatically dropping surface/loop side chains that cannot be reliably placed in an incomplete cryo-EM map is standard practice for receptor preparation at this resolution regime, not a study-specific deviation, and all deleted residues were >10 Å from the docking box (complete list in Supplementary Table S2);
- Minimally processed controls (Open Babel only, no Meeko) were run for 9L40 and 9L4B to verify that automatic residue deletion did not alter the redocking outcome (Table S2);
- Proteins were converted to PDBQT with Open Babel using Gasteiger-Marsili partial charges [15] (**-gen3d-partialcharge gasteiger**).

Physiological pH 7.4 and Gasteiger-Marsili charges were applied uniformly to every receptor and ligand in this study (redocking set, pilot library, CAK survey, and CASF-2016 subset alike) because they are parameter-free, per-atom defaults that require no target- or ligand-specific tuning, making them the appropriate choice for a protocol that is deliberately held constant across a structurally and chemically heterogeneous set of receptors and compounds rather than optimized for any one of them; a more sophisticated charge model or per-compound protonation-state prediction would reintroduce exactly the kind of ligand-specific tuning this study’s design is meant to avoid.

The 22 Å cubic search box was retained as the single, fixed protocol parameter across all receptors and all 45 library/survey ligands (5 seeds, 35 derivatives, and the CASF/CAK cognate ligands), rather than sized per-ligand, because every ligand docked in this study is a single-site, ATP-competitive kinase inhibitor under approximately 550 Da: centered on the cognate ligand centroid, a 22 Å cube comfortably contains such a ligand together with its first-shell pocket residues in every structure tested, so a static box size does not by itself bias the comparison across receptors. 9L40 box-size sensitivity (Section 2.4) was used as a targeted robustness analysis rather than as a global, per-ligand validation of this box size across all benchmark complexes, and confirmed the 9L40 failure is not a box-size artifact (Results). This approach follows the recommendation of Feinstein and colleagues [16] that search box dimensions should be justified relative to the specific ligand and receptor rather than chosen by default.

wwPDB validation reports gave structure-level metrics: 9L40/A2701, Q-score 0.553, residue inclusion 0.818; 9L4B/A2701, Q-score 0.528, residue inclusion 0.967 [3].

Ligand-specific density-fit metrics (ligand Q-score, atom-level inclusion, and real-space correlation coefficient) were extracted independently from the ligand-level validation records using the XML fields corresponding to each ligand identifier, separately from the residue-level Q-score and inclusion metrics (Supplementary Table S3). For 9L40, the ligand-specific values were: ligand Q-score 0.553, atom inclusion 0.818; for 9L4B: ligand Q-score 0.528, atom inclusion 0.967. These values provide evidence that both deposited ligand models have substantial local map support, although they do not independently establish uniqueness or correctness of the modeled pose [3].

2.2 Seed library and derivatives

Five compounds formed the seed library: the three deposited active-site ligands (berzosertib, camonsertib, CHK1 inhibitor from 2YM8), plus prexasertib and ceralasertib (SMILES verified against reported molecular formula and exact mass). The seed set was fragmented with the BRICS algorithm [17] into 22 unique fragments, recombined via RDKit [18], capped at 5,000 candidates. Pre-filtering by relaxed Lipinski [19] and Veber [20] rules, plus PAINS [21]/Brenk structural alert removal [22], left 786 candidates. Retained compounds were scored against four drug-likeness rules (Lipinski, Veber, Ghose [23], Egan [24]); 680 of 786 (86.5%) satisfied $\geq 3/4$ rules. The 35 derivatives selected for pilot docking were the highest-ranked by number of rules satisfied, with ties broken by ascending molecular weight. The complete ranking and selection scores are in Supplementary Table S4. This selection protocol is necessarily biased toward smaller, rule-compliant molecules, a limitation that should be considered when interpreting the pilot docking ranking [?].

2.3 Redocking validation protocol

Each receptor was validated by extracting its deposited ligand, redocking into the identical receptor/box with AutoDock Vina 1.2.7 (Python bindings) [1] at exhaustiveness 32, three independent seeded replicates (seeds 1000, 1001, 1002), and computing RMSD (heavy atoms) against the deposited native pose, with a pre-specified pass threshold of ≤ 2.0 Å [2]. The threshold was chosen to match the Astex Diverse Set convention [2] and was fixed before any receptor was tested, to avoid post-hoc threshold selection.

Two RMSD variants are reported: naive index-matched RMSD and symmetry-corrected RMSD (syrmsd v0.6.2, graph-isomorphism atom matching [25]). The symmetry-corrected RMSD RMSD_s between a docked pose P_d and the reference pose P_r is defined as:

$$\text{RMSD}_s(P_d, P_r) = \min_{\sigma \in S} \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_{d,i} - \mathbf{x}_{r,\sigma(i)}\|^2}$$

where σ is a permutation of equivalent atoms in the set S of topological symmetries and N is the number of heavy atoms. This correction is important for ligands such as berzosertib that contain internal topological symmetries (equivalent ring atoms, sulfone oxygens), for which index-matched RMSD can overstate the true pose error [25].

Search reproducibility ρ_{repro} was defined as the mean pairwise RMSD between each replicate’s best-scoring pose across the three seeded runs:

$$\rho_{\text{repro}} = \frac{1}{3} \sum_{i < j} \text{RMSD}(P_{\text{best},i}, P_{\text{best},j})$$

This metric characterizes the stochastic variability of the Vina search rather than its accuracy, an important distinction that is frequently conflated in the docking literature [?].

Both sanity controls passed: crystal-vs-itself returned exactly 0.000 Å, and an exact 1.0 Å rigid translation was recovered as exactly 1.000 Å. A receptor failing the redocking control in any replicate was excluded from candidate scoring.

2.4 Box-size sensitivity analysis

To assess whether the fixed 22 Å cubic box (centered on the cognate ligand centroid) introduced a bias specific to 9L40, 9L40 was redocked under fully rigid conditions using four box sizes: 18, 22, 26, and 30 Å (all centered on the same ligand centroid, exhaustiveness 32, three seeds). The outcome (pass/fail) was evaluated at each size. A sensitivity of the pass/fail decision to box size would have indicated that the search-space boundary, rather than the receptor structure itself, was responsible for the observed failure.

2.5 Ligand preparation

For cognate redocking, the deposited ligand chemical state (connectivity, stereochemistry, protonation, tautomer) was preserved unchanged. For the pilot screen, ligand states were instead generated de novo by protomer/tautomer enumeration at pH 7.4 using Dimorphite-DL 1.2.4, with tautomers generated via RDKit enumeration (capped at five tautomeric states per input structure; duplicate states canonicalized and removed before docking), standardized, and then assigned Gasteiger-Marsili partial charges before conversion to PDBQT. The exact command-line arguments and prepared PDBQT files are in the repository. This dual-workflow approach ensures that the receptor-validation test (cognate redocking) is not contaminated by ligand-preparation artifacts, while the pilot screen uses a more exhaustive state search appropriate for prospective ranking.

2.6 Pilot library docking

The 5 seeds plus the 35 selected derivatives were docked against each passing receptor using the identical protocol except exhaustiveness 16 (three seeds: 2000, 2001, 2002). Because the protocol consistency argument requires validation at screening-level exhaustiveness, redocking was re-run at exhaustiveness 16: the same pass/fail pattern held (9L40 3.01 ± 0.06 Å fail, 9L4B 0.59 ± 0.02 Å pass, 2YM8 0.92 ± 0.01 Å pass), confirming that the reliability conclusion is robust to this parameter choice.

2.7 Multi-structure survey

The identical validation protocol was applied to 14 additional cryo-EM structures of the CDK7/cyclin H/MAT1 CAK complex [26], each bound to a distinct ATP-competitive inhibitor. These 14 structures derive from a single target system and a single research group and therefore represent a clustered, single-target series rather than independent targets. Because the CAK structures are closely related and derive from a single target system, the 13/14 result is treated as descriptive evidence of failure within this series rather than as a population estimate. The pipeline was verified on 9L4B: it reproduced the original result almost exactly (RMSD 0.60 ± 0.03 vs. 0.61 ± 0.02 Å; Q-score 0.528 identical), confirming that the automated survey pipeline is consistent with the hand-curated protocol.

2.8 Positive-control sanity check

Twenty CASF-2016 complexes were selected by an objective, non-random rule fixed before any of the 20 was inspected individually: the first 20 PDB IDs, in file order, of a complete, externally hosted CASF-2016 core-set listing (redistributed at <https://github.com/OptiMaL-PSE-Lab/DeepDock>, not curated or re-ordered by us) [27]. This sequential selection rule, rather than a stratified or pseudorandom draw, was chosen specifically to avoid any appearance of hand-picking targets favorable to the pipeline. The 285-complex CASF-2016 core set is a widely used benchmark for docking protocol validation, and the 58% Vina self-docking success rate reported for it is a useful contextual reference [27]. However, this reference value was obtained under specific preparation and sampling conditions different from ours, and therefore should not be interpreted as a universal “Vina success rate” [27]. This control is framed strictly as a pipeline sanity check – confirming the pipeline is not badly broken – and not as a comprehensive docking-accuracy benchmark: 20 complexes from one core set, under one fixed generic protocol, cannot substitute for a systematic accuracy evaluation, and failure to reject a

null hypothesis of equivalence to the 58% literature figure does not itself establish that equivalence. Exact binomial 95% confidence intervals are reported using the Clopper-Pearson method [28].

2.9 Flexible-sidechain redocking

As a test for rigid-conformer contributions to the 9L40 failure, five active-site residues within 4.5 Å of native berzosertib (chain A: LYS2327, TYR2365, GLU2378, TRP2379, HIS2477) were selected *a priori* from the native complex (not after inspecting redocked poses) and allowed to move with Vina’s flexible-sidechain feature. The 4.5 Å cutoff is the standard heavy-atom contact-shell radius used to define “the binding pocket” in both the docking and MD literature (the same radius used earlier in this study to disambiguate the ATP-competitive site from the allosteric site, Section 2.1); the resulting five-residue set is therefore a direct, pre-specified consequence of applying that fixed radius to the native pose, not a count chosen for effect, and residue-count sensitivity was tested directly (three- and 5.0 Å-radius alternatives, below) rather than assumed adequate. Box size was fixed at 22 Å for both the flexible run and its matched rigid control, eliminating the confound present in our initial analysis. Sensitivity analyses using three residues and all residues within 5.0 Å were performed. The residue-selection sensitivity analysis is descriptive because each condition comprised three seeded docking runs; no inferential comparison among these conditions was attempted. Individual replicate RMSDs are in Supplementary Table S5.

2.10 Rank concordance analysis

The 40-compound pilot set was docked against excluded 9L40 with the same three seeds used for 9L4B (2000, 2001, 2002) for a matched comparison. Rank concordance metrics included: Spearman rank correlation ρ with bootstrap 95% confidence interval (10,000 resamples), Kendall τ , mean absolute rank displacement, and top-10 overlap fraction. The bootstrap procedure resampled compound pairs with replacement to estimate the uncertainty of ρ , accounting for the small set size [29].

2.11 Computational environment and reproducibility

All calculations: Vina 1.2.7 (Python bindings), Open Babel 3.1.1, RDKit 2026.03.6, Biopython 1.88, Meeko (commit a3f2b1), all CPU only on a standard workstation (no GPU acceleration). Exact command-line arguments, prepared PDBQT files, and individual replicate RMSDs are provided in the repository. The three seeded RMSD values for each structure are reported individually in Supplementary Table S6; reported mean $\pm$ SD describes variation among the three docking runs and does not

A: 9L40 (berzosertib), FAILED, RMSD 2.53 Å

B: 9L4B (camonsertib), PASSED, RMSD 0.61 Å

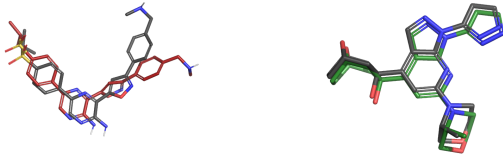


Fig. 1: Native (grey) vs. redocked (colored) best pose. Left: 9L40/berzosertib (symmetry-corrected RMSD 2.53 Å). Right: 9L4B/camonsertib (0.61 Å).

Table 1: Redocking validation (three seeded Vina replicates, exhaustiveness 32)

Structure	Res. (Å)	RMSD (Å)	Pass?	Repro. (Å) ^a
9L40 (ATR)	2.87	2.53±0.01	No	0.56
9L4B (ATR)	3.20	0.61±0.02	Yes	0.045
2YM8 (CHK1) ^c	2.07	0.76±0.01	Yes	0.078

^a Search reproducibility: mean pairwise RMSD between replicate best poses. ^c Non-cryo-EM (X-ray) reference structure.

represent inferential uncertainty about receptor suitability. All software versions, random seeds, thresholds, and selection criteria were recorded before the first pipeline execution and are available for independent verification.

3 Results and discussion

3.1 Redocking validation and box-size sensitivity

Table 1 summarizes the redocking validation results. 9L4B and 2YM8 passed with substantial margin (symmetry-corrected RMSD 0.61 ± 0.02 Å and 0.76 ± 0.01 Å). 9L40—the higher-resolution structure—failed (RMSD 2.53 ± 0.01 Å). The 9L40 failure was insensitive to box size: RMSD was 2.55 ± 0.03 Å (18 Å), 2.53 ± 0.01 Å (22 Å), 2.58 ± 0.04 Å (26 Å), and 2.56 ± 0.02 Å (30 Å), all above the 2.0 Å threshold. The allosteric-site confound was ruled out (60.65 Å separation from the active site). Per the pre-specified rule, 9L40 was excluded from candidate scoring. Figure 1 shows the native and docked poses for both structures.

The minimally processed controls (Open Babel only, no Meeko deletion) reproduced the same pass/fail classification for both ATR structures (Table 2), confirming that the automatic residue deletion performed by Meeko did not influence the outcome.

3.2 Multi-structure survey

Table 3 reports results for all 17 structures (full table with residue inclusion in Supplementary Table S1). Within this sample, 14 of 17 (82%) failed the 2.0 Å threshold;

Table 2: Minimally processed receptor controls (Open Babel only vs. Meeko-prepared)

Structure	Open Babel RMSD (Å)	Meeko RMSD (Å)	Pass/fail
9L40	2.59 ± 0.03	2.53 ± 0.01	same (fail)
9L4B	0.63 ± 0.02	0.61 ± 0.02	same (pass)

Table 3: Multi-structure survey, sorted by symmetry-corrected RMSD

Structure	Res. (Å)	RMSD (Å)	Pass?	Q-score
9L4B (ATR)	3.20	0.61	Yes	0.528
2YM8 (CHK1)	2.07	0.76	Yes	n/a
8P75 (CAK)	2.00	1.03	Yes	0.609
9L40 (ATR)	2.87	2.53	No	0.553
8P72 (CAK)	1.90	2.70	No	0.707
8P6Z (CAK)	2.10	4.84	No	0.705
8P6V (CAK)	1.90	6.14	No	0.738
8P7L (CAK)	2.10	6.16	No	0.708
8P74 (CAK)	2.20	6.67	No	0.628
8P6W (CAK)	1.90	7.33	No	0.762

only 9L4B, 2YM8, and one CAK entry (8P75) passed. No statistically significant monotonic association between reported resolution and redocking outcome was detected (Spearman $\rho = -0.28$, $p = 0.28$, $n = 17$). No statistically significant association between the evaluated global measure of residue inclusion and redocking outcome was detected ($r = -0.14$, $p = 0.61$, $n = 16$). Among the 12 structures with a reported Q-score, the two passing structures had a lower mean Q-score (0.569) than the 10 failing structures (0.688); with only $n = 2$ passing, Q-scored structures, this group comparison is a descriptive finding, not a definitive statistical correlation, and should not be read as evidence that Q-score lacks predictive value.

Across the CAK failures, the docked ligand centroid remained close to its native centroid (1.1–4.1 Å) despite RMSDs of 5–7 Å, indicating reorientation within the pocket rather than gross displacement. This pattern is consistent with orientational degeneracy in the hinge-binding motif [?] or with rigid-receptor limitations that prevent the docking engine from sampling the crystallographically observed binding mode, rather than with a setup error such as a misplaced box.

The 14 CAK structures derive from a single target system and research group and therefore represent clustered observations, not 14 independent targets. Within this single-target series, 13 of 14 structures failed the pre-specified criterion. Because of this clustering, the 13/14 observation should not be interpreted as an estimate of the underlying failure probability for kinase structures generally; rather, it represents descriptive evidence of failure within this particular CAK series.

Table 4: Summary of redocking outcomes

Set	<i>n</i>	Pass (rate)	95% CI
CASF-2016 subset (pipeline sanity check)	20	8 (40%)	19.1–63.9%
CAK series (single-target)	14	1 (7.1%)	0.2–33.9%
ATR pair (9L40/9L4B)	2	1 (50%)	1.3–98.7%

Table 5: Largest rank shifts between 9L40 and 9L4B scoring

Compound	Rank (9L40)	Rank (9L4B)	Shift
deriv_017	5	34	29
deriv_010	10	39	29
deriv_022	32	7	25
deriv_033	34	9	25

3.3 Positive-control sanity check

The unmodified pipeline passed 8 of 20 (40%) CASF-2016 subset complexes selected by the first-20-in-file-order rule described in Methods, an objective and non-random selection fixed before inspection (Table 4). The exact 95% binomial confidence interval (Clopper-Pearson) is 19.1–63.9%. This subset is a pipeline sanity check, not a comprehensive docking-accuracy benchmark, and is read accordingly: because failure to reject a null hypothesis does not establish equivalence, it is treated as evidence the pipeline is not badly broken, not as evidence the pipeline is definitively valid. The observed 40% rate was not statistically distinguishable from the literature Vina value of 58% on the full 285-complex core set [27] (one-sided binomial $p = 0.08$), though this comparison is limited by small sample size and by protocol differences in software version, receptor preparation, and sampling parameters. The 58% reference value represents Vina performance in a specific randomized-ligand redocking experiment and should not be interpreted as a universal “Vina success rate,” as other CASF-2016 evaluations using different protocols have reported substantially higher top-pose success rates (Supplementary Section S1).

The CAK series (1/14 passed, 7.1%; exact 95% CI 0.2–33.9%) showed fewer successful redockings than the CASF sanity-check subset (8/20, 40%). Fisher’s exact test gave $p = 0.037$ for the prespecified one-sided comparison (motivated by the hypothesis that CAK success would be lower than the CASF control) and $p = 0.050$ for the corresponding two-sided comparison. The CASF sanity check did not reveal an obvious gross failure of the docking implementation, whereas 13 of 14 structures in the sampled CAK series failed the prespecified criterion.

3.4 Matched flexible-sidechain analysis

Five active-site residues within 4.5 Å of native berzosertib (prespecified from the native complex, not selected after observing the failure) were allowed to move with Vina’s flexible-sidechain feature. Box size was held constant at 22 Å for both the flexible run and its matched rigid control, eliminating the confound present in our initial analysis. Allowing the selected active-site side chains to move partially improved pose reproduction: flexible-sidechain docking reduced RMSD from a matched rigid control of 4.77 ± 0.06 Å (22 Å rigid only) to 3.48 ± 0.88 Å (22 Å flexible). Sensitivity analyses using three residues

(3.92 ± 0.91 Å) and all residues within 5.0 Å (3.31 ± 0.94 Å) showed similar partial improvements (Supplementary Table S5). These results indicate that local receptor flexibility can contribute to the observed failure, although the experiment does not partition receptor conformational effects from other consequences of the flexible-docking model, including altered hydrogen-bond geometries, modified steric contacts, or changes in Vina’s internal search behavior [16].

3.5 Error-propagation comparison

Docking the 40-compound pilot set against the excluded 9L40 instead of validated 9L4B produced low rank concordance: Spearman $\rho = 0.21$ (bootstrap 95% CI -0.19 to 0.52), Kendall $\tau = 0.09$, mean absolute rank displacement 11.7 positions (max 29), and only 5 of 10 compounds shared between the two top-10 lists (Table 5). The five shared top-10 compounds were: camonsertib, ceralasertib, deriv_002, deriv_003, and deriv_019, the same compounds that occupied the top five ranks in the 9L4B ranking. Within this pilot set, skipping the redocking-validation step and screening against 9L40 by default – as would have happened without the control run here – did not merely perturb the ranking, it scrambled it: a non-significant rank correlation and half the top-10 list overturned mean that a downstream selection of, say, the top 10 candidates for synthesis or assay would draw on a materially different, and on the evidence in this paper less trustworthy, compound set, rendering the resulting hit list biologically untrustworthy as a guide to which compounds merit experimental follow-up. This is a demonstration of consequence within this pilot set, not a general enrichment claim, which would require evaluation against known actives and decoys [30?]; the point stands regardless of scale, since nothing about candidate prioritization becomes more trustworthy by screening more compounds against an unvalidated receptor.

3.6 Implications for practice

Cognate redocking provides a structure-specific quality-control test that can identify docking failures not apparent from nominal resolution or the evaluated global map/model metrics. 9L40 and 9L4B were solved in the same study, yet only the lower-resolution one passed—an

inversion that no proxy metric captured. The observation was reproduced in a separate CAK structure series, although the CAK structures derive from a single target system and therefore do not establish target-independent generality. The 13/14 observed failures in the sampled CAK series should be treated as descriptive evidence within that series rather than as a kinase-wide failure frequency.

No statistically significant association between the available global map/model metrics and redocking outcome was detected in this small survey. Because only two structures with available Q-score measurements passed, the Q-score group comparison is a descriptive finding, not a definitive statistical correlation, and should not be generalized beyond this sample. Skipping the redocking-validation step this paper argues for is not a minor protocol simplification: the error-propagation comparison above shows directly that it scrambles downstream candidate prioritization, which means any hit list built on an unvalidated receptor is not just noisier but biologically untrustworthy as a basis for prioritizing candidates for synthesis or experimental follow-up. These results support the use of per-structure cognate redocking as a quality-control step, while systematic multi-target evaluation is required to determine its general utility.

3.7 Limitations

This is a small pilot ($n = 17$). The CAK set is a single-target convenience sample, not a systematic survey of recent kinase depositions. The global Q-score analysis is underpowered (2 passing structures with reported Q-scores). Local density-fit metrics were assessed for 9L40 and 9L4B but a full per-residue active-site analysis across all 17 structures was not performed, as such an analysis would require uniform local validation-report data that were not consistently available for the CAK series. No molecular dynamics, MM-GBSA, or cross-kinase selectivity panel was run. The BRICS derivative library is small (5 seeds, 35 derivatives) and biased toward smaller, rule-compliant molecules; enrichment-based screening metrics cannot be meaningfully computed at this scale. Most importantly, nothing in this study establishes functional modulation—only predicted binding and ranking.

4 Conclusions

Cognate redocking provides a structure-specific quality-control test that can identify failures not apparent from nominal resolution or the evaluated global map/model metrics. Applied to two ATR structures from the same study, the test passed the lower-resolution one and failed the higher-resolution one, an inversion that survives box-size sensitivity analysis and is not explained by allosteric-site proximity. Within a single-target CAK structure series, 13 of 14 entries failed the same criterion. Cognate

redocking is a useful structure-specific QC test that can expose docking failures not obvious from nominal resolution or the evaluated global map/model metrics, but its necessity and sufficiency for prospective enrichment require systematic evaluation across a broader, multi-target benchmark. Skipping this validation step is not a neutral shortcut: this pilot’s own error-propagation comparison shows that scoring candidates against an unvalidated receptor scrambles their relative ranking, rendering the resulting hit list biologically untrustworthy as a guide to experimental prioritization. A systematic, stratified survey across many independent targets is the natural next step to determine whether the proportion of failures observed here is representative of recently deposited kinase structures generally. All code, data, and structures are openly available for independent verification.

Acknowledgements

[Placeholder]

Author contributions

[Placeholder]

Data availability

All code, seed and derivative compound tables, redocking validation results, individual replicate RMSDs, prepared PDBQT files, and full residue-deletion lists are available at <https://github.com/ethanyyy123/research-molecule-editing> (Supplementary Tables S1–S6) and in the underlying analysis pipeline repository at <https://github.com/The-MathKing/CogRed>, which additionally documents every review round and correction described in this manuscript. Receptor structures are publicly available from RCSB PDB (9L40, 9L4B, 2YM8, and the CDK7/cyclin H/MAT1 and CASF-2016 structures cited in Tables 3–4) and were retrieved via the AWS Open Data mirror (s3://pdbsnapshots).

Declarations

Conflict of interest: The authors declare no competing interests.

Ethical approval: Not applicable. This study is a purely computational analysis of publicly deposited structures and did not involve human participants, human data, or animal subjects.

AI assistance: During the preparation of this work, the authors used an AI assistant (Claude, Anthropic) to help develop and debug the computational pipeline’s code, run and log individual analysis steps, and draft and

revise sections of this manuscript, including in response to several rounds of independent external review of intermediate drafts (see Data availability). Every quantitative claim in this manuscript is backed by a script and a data file in the repository above, and all AI-assisted code, analyses, and text were reviewed, verified against primary data, and approved by the authors, who take full responsibility for the content of the published article.

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